

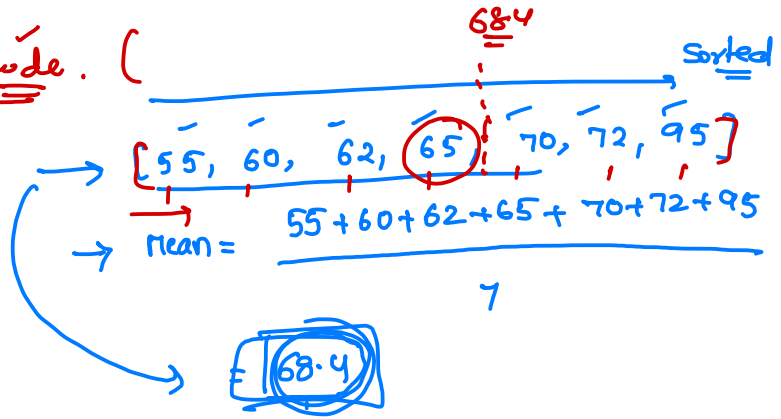


Session	Topics
1.1-1.3	AI Landscape, Lab Setup, Developer Workflow (CLI, Git)
2.1-2.3	Python Basics, Conditionals, Loops
3.1-3.3	Functions, Data Collections, File I/O
4.1-4.3	NumPy, Multidimensional Arrays, JSON
5.1-5.3	Pandas: DataFrames, Selection/Filtering, Cleaning/GroupBy
6.1-6.3	SQL: Tables, Querying, JOINS
7.1-7.3	Matplotlib, Plotly, EDA Checklist
8.1-8.3	APIs, API Security, EDA Workshop
9.1	ML Workflow: Problem Framing, Features vs Labels, Train/Val/Test Split, Baselines
9.2 (Part 1)	Data Prep: Exploring Mess, Text Cleaning, Missing Values, Duplicates, Outliers

- mean / median / mode
- Variance / standard deviation
- gaussian curve
- theory of how model works & distance measures.

→ median (mean / mode) . (

mean ∴ Average



Median: middle value when sorted. → 65

$[55, 60, 62, 65, 70, 72]$

↓

average = $\frac{62 + 65}{2} = \underline{\underline{63.5}}$

Mode: the most frequent / repeated value.

$[55, 60, 62, 65, 70, 72, 90, 72]$

No mode

(72 as mode) .

→ [55 60 62 65 70 72 950] outlier.

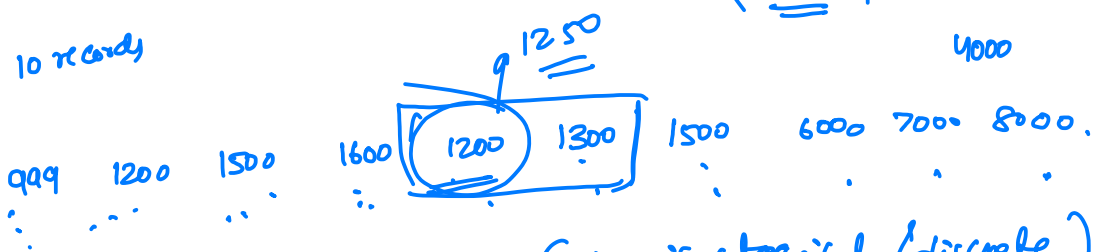
Mean = $\frac{55+60+62+65+70+72+950}{7} = 190.6$ → Mean will get pulled out mostly.

Median = 65 → This is unchanged! (engine) / (mileage)

Mode = No mode

↑
(999cc / 1500cc)

10 records



(Data is categorical / discrete)

5.2

8.1

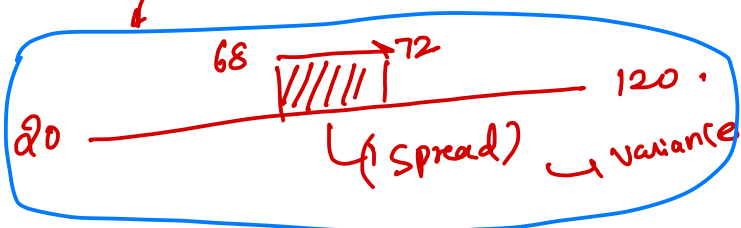
mode imputation

→ Variance / standard deviation

Class A marks: [68 69 70 71 72] → mean = 70 ✓

Class B marks: [20 50 70 90 120] → mean = 70 ✓

all over the place



Data =

Mean =

value

Distance

Square

Variance =

variance (σ^2) =

more 6

(10-6)

$$(96.)^2 = \underline{\underline{100,000}}$$

variance



$$\text{mean} = 6$$

$$s^2 = 8 \text{ (variance)}$$

$$s = 2.83 \text{ (standard deviation)}$$

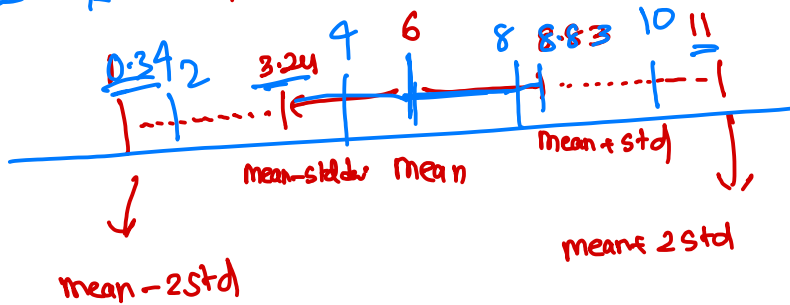
std dev as the

typical distance from mean



$$s = 2.83$$

$$\text{mean} = 6 + 2.83 \Rightarrow 8.83$$

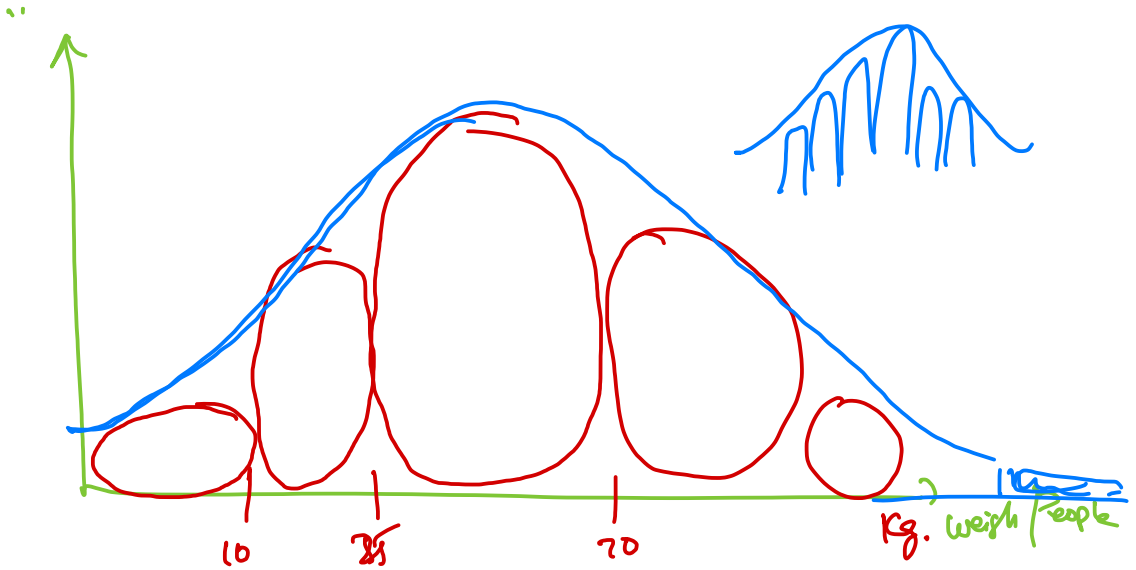
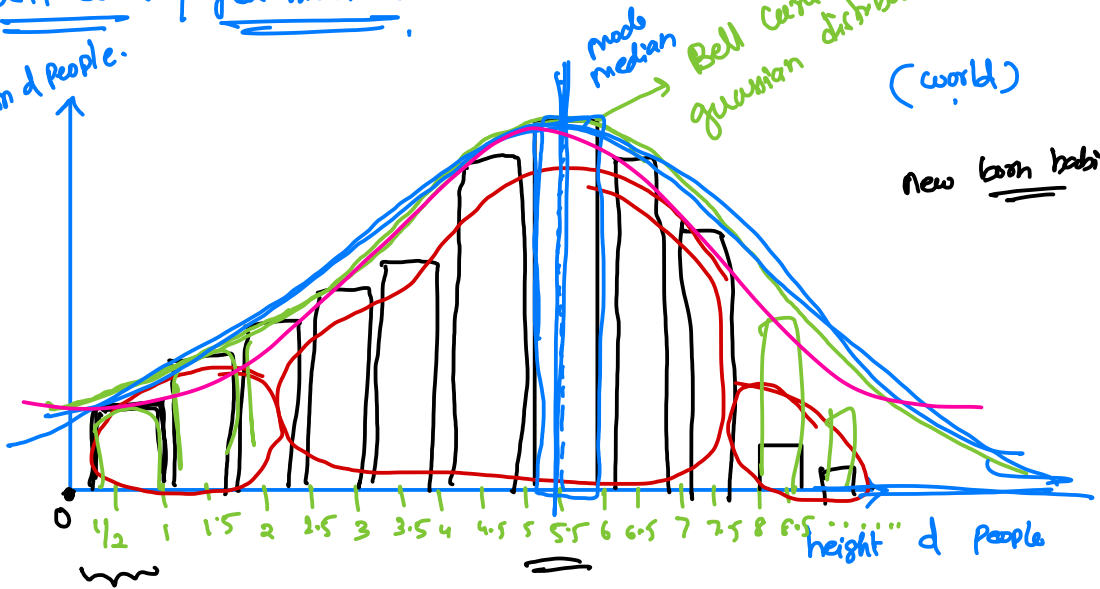


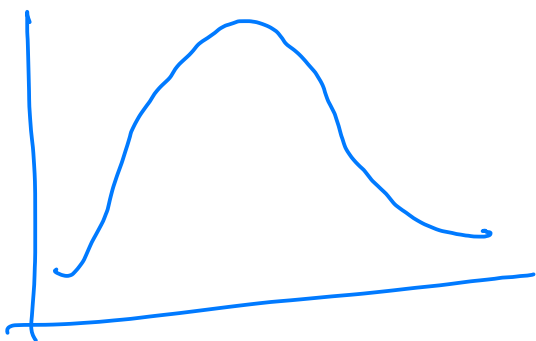
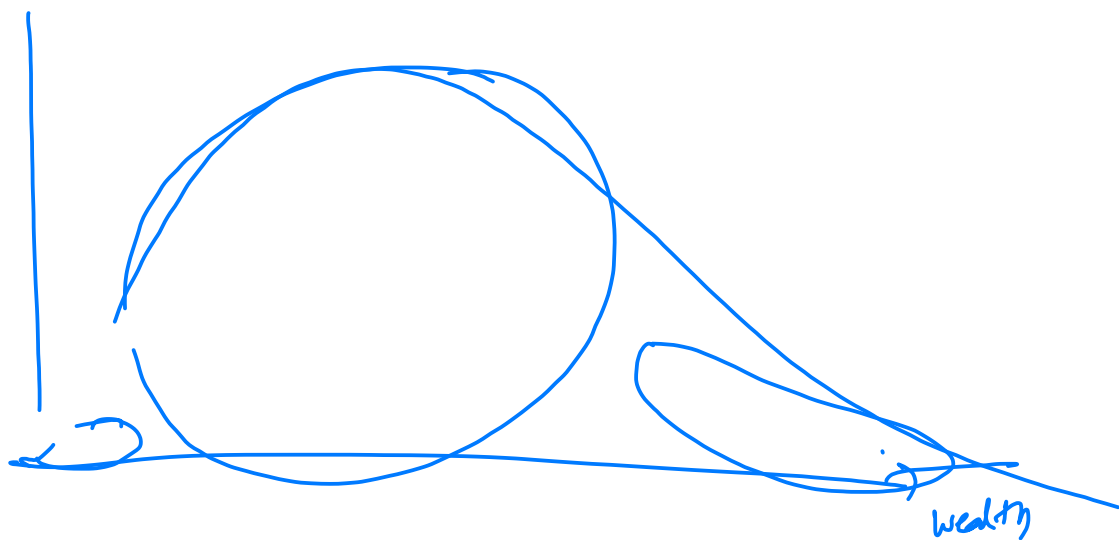
almost all of my data points lie inside

(mean - 2std , mean + 2std)

Bell curve / gaussian curve:

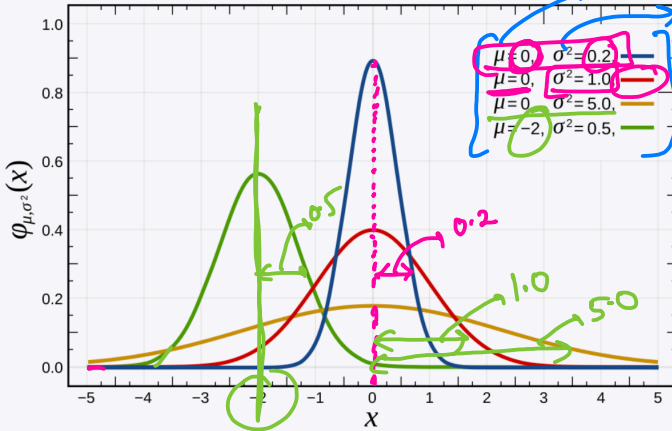
num d people.





Normal distribution

Probability density function



The red curve is the standard normal distribution.

→ Dataset → 1 million records

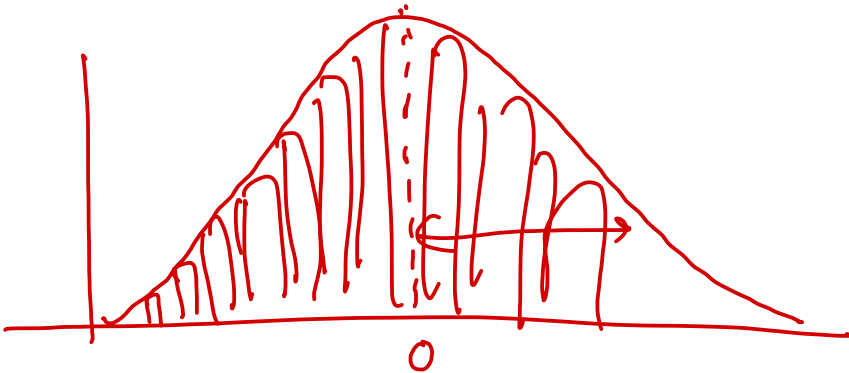
1 column →

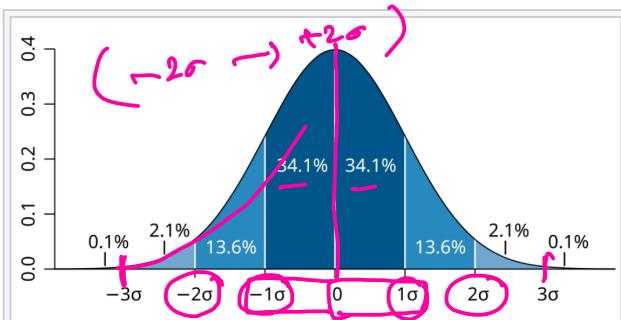
mean

variance

→ it follows

normal dist





For the normal distribution, the values less than one σ standard deviation from the mean account for 68.27% of the set; while two standard deviations from the mean account for 95.45%; and three standard deviations account for 99.73%.

mean = 0

$\sigma = 10$

-10 10
68.2% of data.

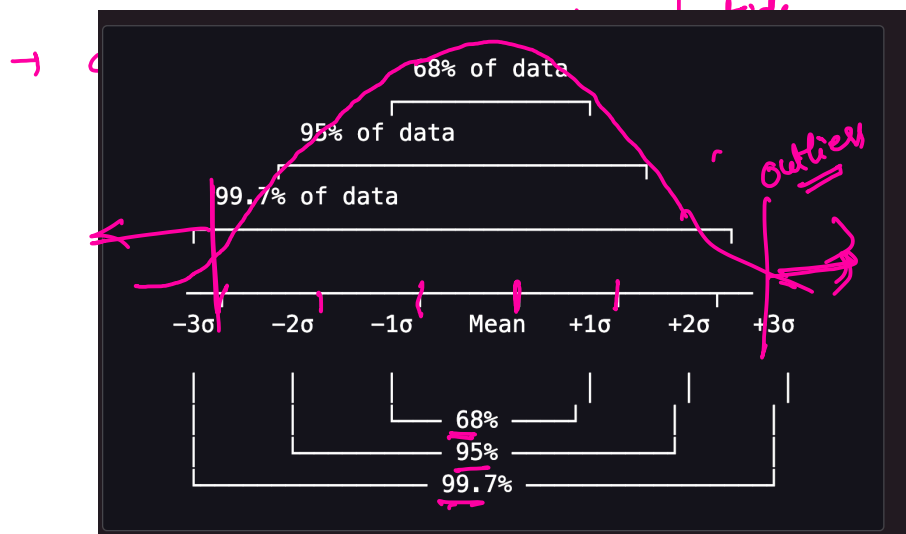
10,000 points.

mean = 0 } $\sigma = 10$
it follows normal dist.

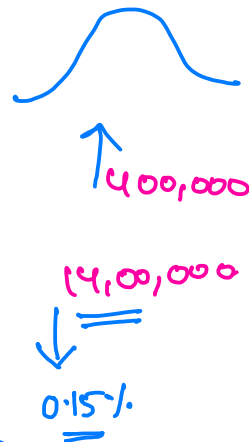
68.2% of these points lie in the range $(-10, 10)$

$(-20, +20) \rightarrow 95.45\%$
 $(-30, +30) \rightarrow 99.73\%$ data.

68.27% $\rightarrow$ -1σ to $+1\sigma$
95.45% $\rightarrow$ -2σ to $+2\sigma$
99.73% $\rightarrow$ -3σ to $+3\sigma$



Selling price: if mean & sp = 500,000
 & std = 300,000



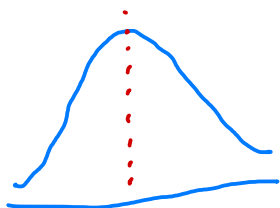
$$\text{mean} - 3\sigma = 500,000 - 3(300,000) \Rightarrow$$

$$\text{mean} + 3\sigma = 500,000 + 3(300,000) \Rightarrow$$

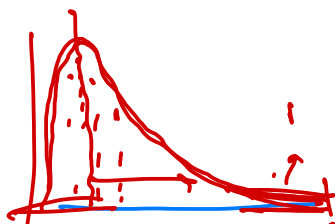
(outlier detection)

99999999 $\Rightarrow$ 300
std away

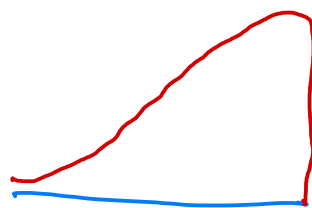
$\rightarrow$ Not all everything is Normal $\rightarrow$ ok



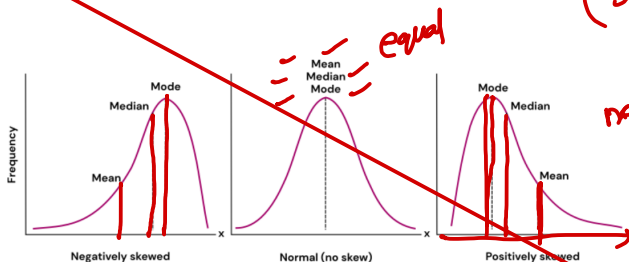
$\downarrow$
 Normal



$\downarrow$
 house prices / deposits



(skewness)



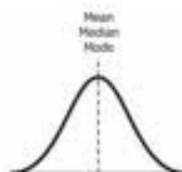
mean > median > mode

(mean < median < mode)



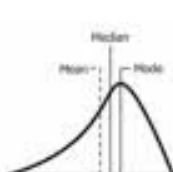
FLASHCARD

**Normal
Distribution**



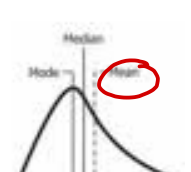
$\text{Mean} = \text{Median} = \text{Mode}$

**Left-Skewed
Distribution**



$\text{Mean} < \text{Median} < \text{Mode}$

**Right-Skewed
Distribution**



$\text{Mean} > \text{Median} > \text{Mode}$

PSM by Kashish Grover

Facebook Instagram Twitter @kays_psm



Dr. Kashish Grover
MBBS, MD (PSM)
PSM Faculty

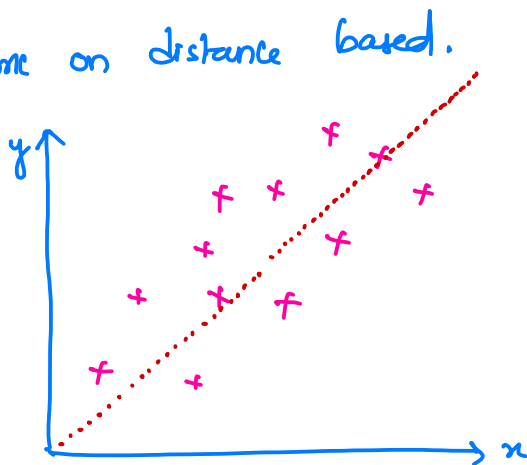
→ Scaling of the data:

↳ most of the ML model work on distance based.

100,000
150,000
300,000
500,000

→ Scatter plot

↑
Predict this one

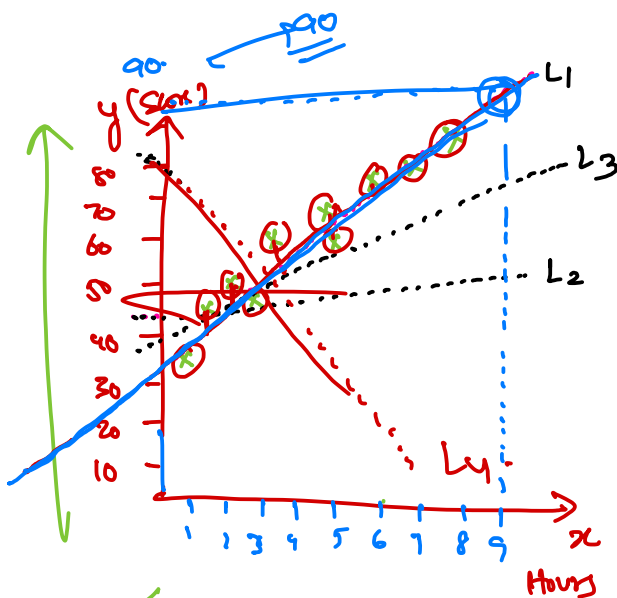


Point	Hours (x)	Score (y)
1	1	35
2	2	45
3	2.5	50
4	3	48
5	4	60
6	5	65
7	5.5	62
8	6	70
9	7	78
10	8	85

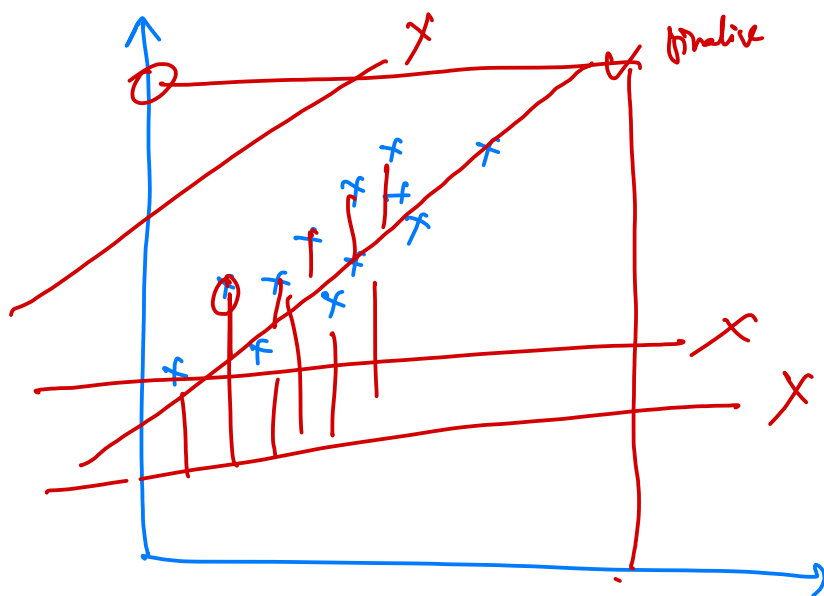
⑨ → ? Target

(all from)

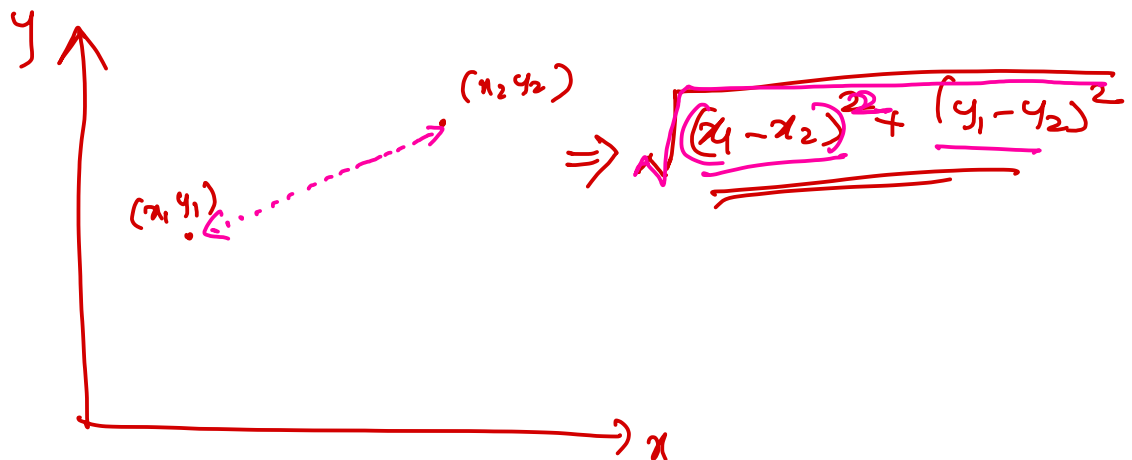
10 hours?



→ ML will try to find a line (curve) that minimises the distance from all the points to the line?



$$\underline{\underline{y = mx + c}}$$



Car price :

Car age

| km driven.

①

$$\left\{ \begin{array}{l} \text{Car age} = 54 \\ \text{Car age} = 39 \end{array} \right\}$$

$$\begin{array}{l} \text{km driven} = 50,000 \text{ km} \\ \text{km driven} = 30,000 \text{ km} \end{array}$$

②

Distance $\Rightarrow$ $(3-5)^2 + (50,000-30,000)^2$

$\Rightarrow$ $29,000$

(x_1, y_1)

(x_2, y_2)

$(-2)^2 = 4$

$400,000,000$

$(100,000,000 \div 1)$ variance

is age really 100,000,000 times less important compared to km driven? (No) (will also think like that)

$\rightarrow$ solution is scaling?

Scaled value = $\left[\frac{x - \text{mean}}{\text{std.}} \right]$

km_driven
↓
mean
↓
70,000
std = 40,000

Car age
↓
mean
↓
7
std = 4

$(2)^2 = 4$

Car age (0-1)

5	50,000	0.5
3	30,000	1

$(0.5)^2 = 0.25$
 $(1)^2 = 1$
 0.25

$\frac{(50,000 - 70,000)}{40,000}$

$\left(\frac{30,000 - 70,000}{40,000} \right)$

→ 0.4	0.5
1 1	-1
-	

(calculate distance)
Balay

$(0-1)$

$\sqrt{(0.4-1)^2 + (0.5-1)^2}$
 $= \underline{(0.6)^2} + \underline{(0.5)^2}$